VectorCube Construction Manual

Overview

The VectorCube is a modular active (airflow-based) insect trap designed to attract and capture blood-searching insects using controlled airflow, visual attraction, and CO_2 baiting. The system consists of a central airflow distributor, an inline duct fan, a pipe-based intake structure, and a supporting frame.

Air is drawn through four horizontal and one vertical intake arms into a central block and then pulled upward by a fan, creating directional flow that guides insects into the trap.

Functional Principle

Airflow generation

A central fan generates suction, drawing air through four horizontal and one vertical intake pipes.

Directional guidance

The interior block distributes airflow evenly, ensuring consistent suction from all directions.

Vector attraction

A CO_2 source (e.g., yeast fermentation, bottled CO_2, or dry ice) can be used to attract insects and guide them into the airflow. Additionally, the colored tarpaulin provides visual attraction.

Vector capture

Different capture methods can be implemented:
1. A capture net placed after the fan (captures are damaged)
2. A capture chamber positioned before the fan (under development)
3. A mesh placed at the fan intake combined with a capture chamber connected to one of the horizontal intake pipes

Components

1. Interior Airflow Block

Files:
VectorCube_interior_block_milled.png
VectorCube_interior_block_printed.png

Central cube that connects all pipes and the fan. Contains internal channels to distribute airflow. Can be manufactured via CNC milling or 3D printing. Ensure smooth internal geometry and airtight pipe interfaces.

2. Fan Adapter (only for milled internal block)

File:
vectorcube_fan_mount_detail_milled.jpg

Milled connector between fan and interior block. Optionally, duct tape can be applied around the fan can replace adapter. Ensures airtight transition and proper alignment. The ridges on the fan housing were sanded down to ensure a smooth, airtight fit with the adapter.

3. Pipe System

Files:
vectorcube_interior_system_assembled.jpg
vectorcube_interior_system_assembled2.png

Four horizontal and one vertical intake pipes.
Material: DN 75 wastewater pipes with integrated push-fit sockets (European standard).

4. Inline Duct Fan

File:
inline_duct_fan_spec.png

Diameter: 100–125 mm
Airflow: approx. 270 CFM
Power: typically 12V

Generates the suction force driving the airflow through the system. Can be used with an adjustable power supply for airflow adjustment.
An 87° DN 110 pipe elbow was installed at the outlet to prevent rain from entering the system.

5. Frame Structure

Files:
VectorCube_frame_front.png
VectorCube_frame_and_interior_front.png

Inner support structure defining the geometry of the trap and holding airflow system in place.
Material: 20 × 20 mm aluminium square tubing with compatible plastic corner connectors.

6. Openings / Inlets

File:
VectorCube_openings_top.png

Precisely positioned entry points aligned with intake pipes to control how insects enter the system.

7. Capture Chamber

Components:
- 75 mm to 50 mm reducer (HT pipe)
- 50 ml Falcon tube

The capture chamber can be connected to one of the horizontal intake pipes. The Falcon tube serves as a removable collection container.

Cover

The frame is enclosed by a custom-made tarpaulin cover. The enclosure includes a lid that can be closed using hook-and-loop fasteners. Colored foils can be attached to the exterior using the same fastening system to provide visual attraction.

The cover is equipped with two attachment loops, allowing a rope or strap to be fastened so the device can be carried over the shoulder or suspended during operation.

Assembly Instructions

1. Prepare Interior Block: Manufacture or 3D print the airflow block. Ensure internal channels are clean and unobstructed.
2. Attach Fan: Mount the fan on top of the block (use adapter if milled). Ensure airtight connection.
3. Install Pipes: Insert horizontal and vertical pipes. Ensure tight fits and seal if necessary.
4. Mount Fan: Verify airflow direction (must pull air upward).
5. Assemble Frame: Construct frame and position system centrally.
6. Align Openings: Ensure intake openings align with pipe ends an push them through slightly for stabilisation.
7. Final Assembly: Secure all components and check stability.

Electrical Setup

Connect the fan to a suitable power supply (e.g., 12V DC). Ensure correct polarity. Optional: switch or speed controller.

Optional CO_2 System

Use a mixture of sugar, yeast, and water. Route CO_2 output into the airflow region and position near intake.

Final Checklist

- All connections are airtight
- Fan airflow direction is correct
- Pipes are securely mounted
- Frame is stable
- No internal blockages
- Electrical connections are safe